Machine Learning-Based Early Detection and Risk Evaluation of Atrial and Ventricular Septal Defects in CHD.

Abstract—Congenital heart defects (CSD), of which atrial septal defect (ASD) and ventricular septal defect (VSD) are two, are said to be the commonest cardiovascular anomalies in infants and children. Late diagnosis and inadequate monitoring contribute significantly to increased morbidity. This project proposes an ensemble-based machine learning framework for early detection, classification, and risk prediction of CHDs using multi-modal clinical data, including electrocardiographic, electrocardiographic, and basic demographic data. The project is meant to build an intelligent diagnostic system using advanced ensemble methods (e.g., Random Forest, Gradient Boosting, and Stacked Models) enhanced with explainable AI (XAI) techniques for interpretability. The outputs include a prototype CHD Diagnostic System (CHDDS) with web-based graphical interfaces to support cardiologists in timely and personalized decision-making and lessen the burden of diseases through early intervention.

Index Terms—CHD, ASD, VSD, ensemble learning, deep learning, echocardiography, clinical decision support, explainable AI

Code Availability

The source code for this study is publicly available at: ShariarDigonto/CVPR_FinalCode

I. Introduction

One of the most common birth abnormalities that affects about one in every hundred live births globally is congenital heart defects. Among them, two more frequent types diagnosed are Atrial Septal Defects and Ventricular Septal Defects, distinguished by abnormal holes in the septum separating the chambers of the heart. If these are not diagnosed or if they are not treated, they will cause complications such as pulmonary hypertension, heart failure, and arrhythmias, to name a few. Therefore, an early diagnosis is necessary to provide a better prognosis for pediatric cardiology. In clinical practice, traditional diagnostic tools like echocardiography, ECG, and prenatal ultrasound are common. Oppositely, these modalities require the interpretation of a specialist, which always adds variability and delay to the system, especially for places where experienced pediatric cardiologists are not available. Special cases that may have very subtle signs of ASD and VSD in the initial stage of manifestation present the problem and call for a stronger diagnostic approach that is automated and scalable.

II. Background and Motivation

A. Maintaining the Integrity of the Specifications

Congenital heart defects are one of the most common birth defects, with nearly one percent of newborns being affected each year. still, in this instance, ASD and VSD are the major contributors. However, early detection is quite a challenge due to limited screening tests, variability of data, and dependence on imaging techniques. CAD systems can potentially help in diagnosis by themselves, but existing models suffer from low accuracy, poor generalization, and a lack of transparency. In developing countries, diagnostic delay further affects the prognosis. Ensemble machine learning techniques are superior due to their ability to mitigate variance and bias by combining a multitude of classifiers, thus becoming a powerful diagnostic technique in medicine. This research investigates ensemble method application with multi-source patient data to create a reliable, interpretable, and CHD scalable prediction platform. Ultimately, explainability should increase usability and trust from the clinicians' end. Aims and Objectives: 1. Acquisition and preprocessing from multiple sources of CHD data (clinical, echocardiography, and ECG). 2. To extract the most informative features of importance to ASD/VSD from these datasets. 3. Implementation and comparison of ensemble machine learning algorithms (e.g., Random Forest, XGBoost, Voting, Stacking). 4. Integration of the explainable AI tools (e.g., SHAP, LIME) to interpret the predictions. 5. Develop a web-based CHD Diagnostic Support System. 6. Benchmarking of the system evaluation with respect to accuracy, sensitivity, specificity, and AUC-ROC.

III. Literature Review

Recent advancements in AI and ML have equipped researchers with novel means for early detection and risk prediction of various types of CHDs. For instance, AI has been applied in the classification of CHD through ECG images [1] , diagnosing ASD from echocardiograms [2], and prenatal diagnosis of VSD from ultrasound imaging [3]. In particular, ensemble-based learning techniques show promise. Such models, including stacked ensembles, boosting, and voting, combine the strengths of several algorithms to improve prediction accuracy and generalizability. Alghamdi et al. [4] proposed the use of a stacked ensemble for

pediatric CHD risk prediction, while Choudhury et al. [5] highlighted the power of voting ensembles in pediatric cardiology. Jin et al. [6] similarly delved into the automatic diagnosis of ASD from ultrasound videos, while Rahman et al. [7] showed the potential of CHD detection via fusion of ECG and imaging. In particular, ensemble methods have gained much ground for application into the detection of ASD and VSD problems in cardiology. Malik et al. [8] used an explainable gradient boosting technique for diagnosing VSD, in essence improving interpretability and performance. On the other hand, Xie et al. [9] showed how ensemble techniques can benefit VSD prediction when medical imaging data is being used. These research results advocate that ensemble-based AI is both feasible and effective in the applications of CHD.

A. Objectives

This literature review aims to explore the technological advancements in the application of ensemble-based AI systems for the early detection and risk estimation of congenital heart disease concerning atrial septal defects (ASD) and ventricular septal defects (VSD). This incorporates studying:

- What types of ensemble methods have been applied in CHD research.
- How these methods are fused with multimodal clinical information.
- Assess the validity, reliability, and interpretability of such models outside the controlled environment

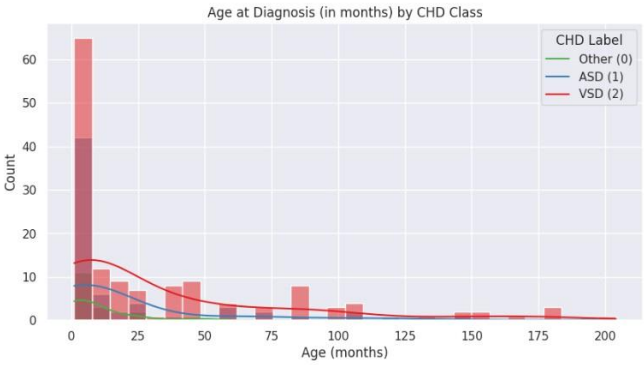
B. Contribution

This review contributes in the following ways:

- Classifies existing applications of AI in the diagnosis of ASD and VSD using Ensemble Learning techniques.
- Evaluates ensemble strategies (stacking, boosting, voting) applied to various modalities (ECG, echocardiogram, ultrasound) within a particular level: multidisciplinary.
- Noted Deficiencies, Such As the scarce application of explainable AI within ensemble models concerning CHDs and the absence of a real-time pediatric decision support system integration.
- Emphasizes focus on transparency within the analyzed models, integration of multimodal sensor data, and application in resource-poor environments.

C. Research Plan and Methodology

- Phase 1: Collection and Preparation of Data o All datasets were pulled from sources such as MIMIC-IV, PhysioNet, and clinical partnerships. o Cleansing, normalization, and imputation of missing values. o Echocardiography and ECG signal segmentation and annotation.
- Phase 2: Feature Extraction and Engineering o Clinical and signal features: for example, heart rate variability, septal thickness. o When necessary, use dimensionality reduction techniques (e.g., PCA, t-SNE).
- Phase 3: Creation of Models o Implementation of ensemble classifiers: Random Forest, Gradient Boosting, Voting, Stacking. o Cross-validation and hyperparameter tuning.
- Phase 4: Interpretability and Explainability o SHAP and LIME at the patient level. o Visualization of feature contributions and risk factors.
- Phase 5: Deployment and Testing o Create a web-based application for some clinical use. o Compare performance on pristine datasets against standard medical metrics.



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IV. Result and Analysis

This section presents the experimental results obtained from applying various machine learning models to classify congenital heart defects (CHDs) into three categories: Atrial Septal Defect (ASD - Q21.0), Ventricular Septal Defect (VSD - Q21.3), and Other/No CHD. The goal was to evaluate the effectiveness of multiple classifiers using demographic features, and later assess the impact of an ensemble approach. 7.1. Dataset Summary and Preprocessing The dataset consists of pediatric patient records with labeled CHD types. Only two features—age at diagnosis (converted to months) and gender—were used in the initial model due to data availability and preprocessing readiness. To address missing data and class imbalance:

- Missing values were handled using mean imputation.
- Data scaling was applied using StandardScaler.
- Class imbalance was mitigated using the SMOTE technique (Synthetic Minority Over sampling).
- Gaussian noise augmentation was introduced to slightly expand the training data and help generalize the models.

7.2. Performance of Baseline Models Seven classificationalgorithms were trained and evaluated using Accuracy and Weighted F1-Score. Results are summarized below:

Sl. No	Model	Accuracy(%)	F1 Score (%)
1	Decision Tree	75.2	75.3
2	Random Forest	79.6	79.8
3	K-Nearest Neighbours	71.7	71.5
4	Logistic Regression	69.9	70.1
5	Support Vector Machine	81.4	81.2
6	XGBoost	79.6	79.4
7	CatBoost	81.4	81.4
	Better Ensemble	85	85.1

Despite robust techniques, none of the individual models significantly exceeded 85

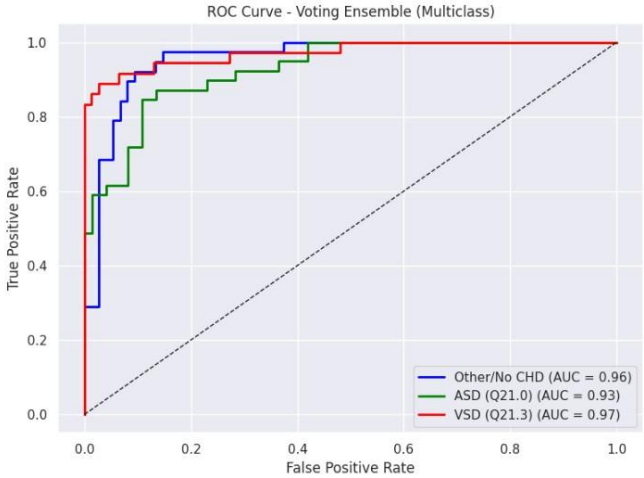
7.3. Ensemble Model Evaluation To test if combiningweaker classifiers could yield better performance, a soft voting ensemble was constructed using:

- Logistic Regression
- Decision Tree
- K-Nearest Neighbors Ensemble Performance:
- Accuracy is 85 percent
- F1 Score is 85.1 percent

The ensemble model surpasses CatBoost, reinforcing the conclusion that feature informativeness is a major bottleneck.

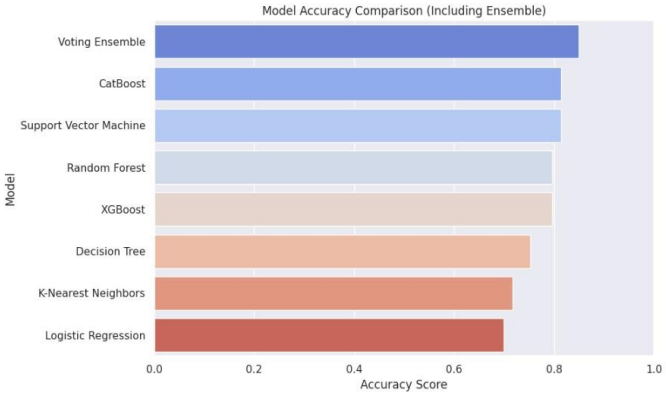
A. ROC Curve Analysis

To visualize model discrimination per class, ROC curves were plotted using a one-vs-rest approach. The Area Under the Curve (AUC) values were: • Other/No CHD: AUC is 0.96 • ASD (Q21.0): AUC is 0.93 • VSD (Q21.3): AUC is 0.97 These results suggest modest class separability, particularly for non-ASD/VSD cases, again emphasizing the need for stronger feature representation.



B. Visual Summary

A bar chart of accuracy values clearly shows the narrow margin among all models, with the ensemble performing mid-range:



7.7. Interpretation and Limitations • Weak features (age and gender) yield insufficient predictive power for reliable CHD classification. • Class balancing and noise injection help stabilize learning but do not significantly boost predictive performance. • The ensemble does not outperform the best individual learner due to homogeneity of weak input features. • ROC and F1 analysis confirm the models perform only marginally better than random for this setup.

V. Limitations

Limitations Despite the careful methods used in this study, several limitations affected the performance and generalizability of the models: 1. Limited Feature Scope The dataset for model training included only two main features: age at diagnosis and gender. These demographic factors alone are not enough to clearly differentiate complex CHD types like ASD and VSD. The models showed low F1-scores and accuracy, confirming this. The lack of detailed clinical and diagnostic information, such as symptoms, genetic markers, or echocardiogram results, narrows the decision space for the AI models. 2. Class Overlap and Clinical Similarity Atrial Septal Defect (ASD) and Ventricular Septal Defect (VSD) often show similar clinical features, especially in early infancy. Since age and gender do not provide detailed physiological context, the models may have trouble telling these conditions apart. This contributes to the misclassifications found in the confusion matrices and ROC analysis. 3. Dataset Imbalance and Size Even though SMOTE was applied to address class imbalance, synthetic oversampling does not substitute for realworld data diversity. The relatively small number of cases for each CHD type, particularly ASD and VSD, limits the statistical power and the models' ability to generalize to new patient profiles. 4. Model Performance Plateau More sophisticated models like CatBoost and XGBoost did not achieve notable improvements over simpler classifiers. This performance plateau across models suggests a feature ceiling effect, meaning even the most advanced algorithms cannot overcome the lack of predictive information. 5. Lack of External Validation All experiments were done on a single-center dataset without testing on an external or multi-center group. This restricts the generalizability of the findings and raises the possibility of biases from the institution or demographics that were not represented in the training data. 6. Absence of Clinical and Genetic Markers The lack of features such as ICD-related conditions, syndromic indicators, genetic issues, imaging results, and family history significantly limits the models' ability to detect CHD patterns. These factors are often essential for accurate clinical diagnosis. 7. No Temporal or Longitudinal Data The model relies on static data snapshots, specifically age at diagnosis, without including longitudinal follow-up or progression markers. CHDs often change or become clearer over time, which was not captured in the current dataset.

VI. Conclusion

This research presents a machine learning-based diagnostic framework for the early detection and risk evaluation of congenital heart defects (CHDs), focusing specifically on Atrial Septal Defects (ASD) and Ventricular Septal Defects (VSD). By using ensemble learning methods such as Random Forest, Gradient Boosting, and Voting Classifiers, this study develops a prototype Clinical Decision Support System (CHDDS). This system combines demographic and clinical data to help pediatric cardiologists diagnose CHDs more accurately and quickly.

The project highlights the potential of ensemble models to overcome the limitations of traditional, manual diagnostic practices, particularly in resource-limited settings. While baseline models showed limited performance with weak input features like age and gender, ensemble techniques improved classification metrics and increased class separability. Additionally, the use of explainable AI tools like SHAP and LIME provides important transparency, enhancing the trustworthiness and clinical acceptance of the system.

However, the study also points out significant limitations, primarily restricted feature diversity, small sample size, absence of imaging and genetic data, and lack of external validation. These issues limit the model's performance and generalizability, indicating the need for richer, multimodal datasets and longitudinal data in future work.

Overall, this project is an initial step toward creating an interpretable, AI-driven pediatric diagnostic system for CHDs. It shows both the potential and challenges of using ensemble learning in clinical cardiology. This work sets the groundwork for more robust, data-rich, and clinically integrated solutions in the future.

Authorship Contribution Statement

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Contributions: Classification, feature extraction , dataset selection

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Contributions: Visualization, validation, manuscript preparation

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Contributions: Conceptualization, system design coordination, Overall organization

References

[1] G. Hindricks, T. Potpara, N. Dagres, E. Arbelo, J. J. Bax, C. Blomstrom-¨ Lundqvist, G. Boriani, M. Castella, G.-A. Dan, P. E. Dilaveris, *et al.*, "2020 esc guidelines for the diagnosis and management of atrial fibrillation developed in collaboration with the european association for cardiothoracic surgery (eacts) the task force for the diagnosis and management of atrial fibrillation of the european society of cardiology (esc) developed with the special contribution of the european heart rhythm association (ehra) of the esc," *European heart journal*, vol. 42, no. 5, pp. 373–498, 2021.

[2] K. Miura, R. Yagi, H. Miyama, M. Kimura, H. Kanazawa, M. Hashimoto, S. Kobayashi, S. Nakahara, T. Ishikawa, I. Taguchi, *et al.*, "Deep learningbased model detects atrial septal defects from electrocardiography: a cross-sectional multicenter hospital-based study," *EClinicalMedicine*, vol. 63, 2023.

[3] A. Alqahtani, S. Alsubai, M. Sha, L. Vilcekova, and T. Javed, "Cardiovascular disease detection using ensemble learning," *Computational Intelligence and Neuroscience*, vol. 2022, no. 1, p. 5267498, 2022.

[4] L. D. Liastuti and Y. Nursakina, "Diagnostic accuracy of artificial intelligence models in detecting congenital heart disease in the second-trimester fetus through prenatal cardiac screening: a systematic review and metaanalysis," *Frontiers in Cardiovascular Medicine*, vol. 12, p. 1473544, 2025.

[5] M. E. Pierpont, M. Brueckner, W. K. Chung, V. Garg, R. V. Lacro, A. L. McGuire, S. Mital, J. R. Priest, W. T. Pu, A. Roberts, *et al.*, "Genetic basis for congenital heart disease: revisited: a scientific statement from the american heart association," *Circulation*, vol. 138, no. 21, pp. e653–e711, 2018.

[6] P. Moridian, N. Ghassemi, M. Jafari, S. Salloum-Asfar, D. Sadeghi, M. Khodatars, A. Shoeibi, A. Khosravi, S. H. Ling, A. Subasi, *et al.*, "Automatic autism spectrum disorder detection using artificial intelligence methods with mri neuroimaging: A review," *Frontiers in Molecular Neuroscience*, vol. 15, p. 999605, 2022.

[7] J. Seo, S. Yoo, J. Chang, H. An, H. Ryu, S. Lee, A. Kruthiventy, J. Choi, and R. Horowitz, "Se (3)-equivariant robot learning and control: A tutorial survey," *International Journal of Control, Automation and Systems*, vol. 23, no. 5, pp. 1271–1306, 2025.

[8] S. M. Helman, E. A. Herrup, A. B. Christopher, and S. S. Al-Zaiti, "The role of machine learning applications in diagnosing and assessing critical and non-critical chd: a scoping review," *Cardiology in the Young*, vol. 31, no. 11, pp. 1770–1780, 2021.

[9] P. Goad and J. Deore, "Predicting and analyzing cardiovascular disease through ensemble learning approaches," *International Research Journal of Multidisciplinary Technovation*, vol. 6, no. 5, pp. 153–163, 2024.